

When to Move On: What Self-Play Agents Learn About Commitment Under Competition

A simulation study of optimal stopping in congested choice —
shoe shopping and dating as running examples

September 2026

Simulation engine, interactive lab, and all experiment data are released alongside this paper.

Abstract

When you shop for shoes — or date — you face an *optimal stopping* problem: each option you inspect costs time, and the good ones may be gone tomorrow. The classic answer, the secretary problem's 37% rule, assumes no competition and no second chances. We study what changes when options are *scarce* and *rivals want the same ones you do*.

We build a simulated market in which N agents shop M shoe styles over T time steps. Each style has limited stock and each agent's satisfaction is a mix of common popularity and idiosyncratic taste, $v = w \cdot \text{popularity} + (1-w) \cdot \text{taste}$. Agents learn a threshold strategy — browse for a while, then buy the first option clearing a bar that *falls* as the deadline approaches — via self-play evolution. We train agents across 27 market configurations varying competition, stock scarcity, the satisfaction distribution, taste agreement, time budget, a dating-market variant, and a safety-net variant.

Four findings stand out. **First**, every learned strategy starts ultra-picky (threshold 1.0) and lets the *declining bar* do the exploring; the closing-time bar is the single adaptive dial that encodes competition, scarcity, and deadlines. **Second**, rigid heuristics break exactly when markets get hard: the 37% rule loses by 0.09 under single-unit stock and by 0.34 under maximum competition, because it anchors on options rivals remove mid-search. **Third**, welfare is *U-shaped* in taste agreement — partial agreement is worse than none or total. **Fourth**, in the dating variant the learned strategy is "pounce on love at first sight, then lower your standards fast" (no browsing, bar $1.00 \rightarrow 0.06$); even random committing beats the 37% rule there. The practical answer to "when to move on": commit when something clears a bar you lower as time passes — but in a scramble, never explore passively.

1 Introduction: the question

You want a pair of shoes. The mall has dozens of styles; each has a satisfaction score only you can judge by trying it on. Trying is free but takes time, the mall closes at nine, and other shoppers may buy the last pair in your size while you dither. When do you stop looking and commit? Dating has the same shape: each person you meet costs weeks, and the wonderful ones pair off while you "keep your options open."

Decision theory has a canonical answer for the *uncongested* case: the secretary problem. Interview 37% of candidates with no offers, then hire the first one better than all seen so far — optimal when candidates are ranked, all are seen, and nobody else is hiring (Ferguson, 1989; Gilbert and Mosteller, 1966). Real markets violate all three assumptions: satisfaction is cardinal, inspection consumes the budget, and rivals deplete the stock. This paper asks what *learned* strategies look like once competition is real — not derived analytically, but discovered by agents that must survive the market.

Our contribution is a compact, reproducible laboratory: a numpy-only market simulator, a self-play evolution trainer, 27 trained configurations, an interactive Streamlit lab, and the behavioral regularities the agents discovered. The headline is a simple decision rule with one adaptive dial, plus a set of precise statements about when textbook rules fail.

2 The market model

2.1 Shoes, satisfaction, and stock

N agents shop M shoe styles over T discrete time steps. Style j has a popularity p_j (common value: how good it is "objectively") and a stock of s_j identical pairs. Agent i 's satisfaction from style j is

$$v_{ij} = w \cdot p_j + (1 - w) \cdot t_{ij}, \text{ clipped to } [0, 1]$$

where t_{ij} is idiosyncratic taste drawn i.i.d. from a configurable distribution. The parameter $w \in [0, 1]$ is *taste agreement*: $w = 1$ is a pure popularity contest (everyone agrees what's good); $w = 0$ is purely personal taste (your favorites are nobody else's business). Both p and t are drawn from one of four distributions — uniform, normal, beta(2,5) (most options "meh"), or a heavy-tailed analog (rare gems) — letting us ask how the *shape* of satisfaction changes optimal pickiness.

2.2 The shopping protocol

- Each step, agents act one at a time in a fresh random order.
- **Inspect**: reveal your own v for the next unseen style (each agent samples styles in a private random order — sampling without replacement). Costs the turn.
- **Buy**: take one copy of any previously inspected style with stock > 0 . You exit immediately with payoff v .
- **Pass**: do nothing; keep searching.
- A purchase decrements stock; at zero the style is gone for everyone. Remaining stock is observable, so agents never waste an inspection on a sold-out style. If the deadline arrives with no purchase, payoff is 0 ("go home empty-handed") — except in the safety-net variant, where the agent is forced to take its best inspected, still-available style.

2.3 What the agents can learn

Agents learn a *threshold strategy* with three parameters (e, θ_0, θ_1): inspect-only for the first e turns ("browse"), then each turn inspect the next style and buy the best inspected style still in stock if its value $\geq \theta(t)$, where θ declines linearly from θ_0 to θ_1 over the trip. This class nests "the 37%-style rule" (long browse, high bar) and "grab early" (short browse, low bar). Figure 1 shows a learned schedule: the falling bar *is* the exploration mechanism.

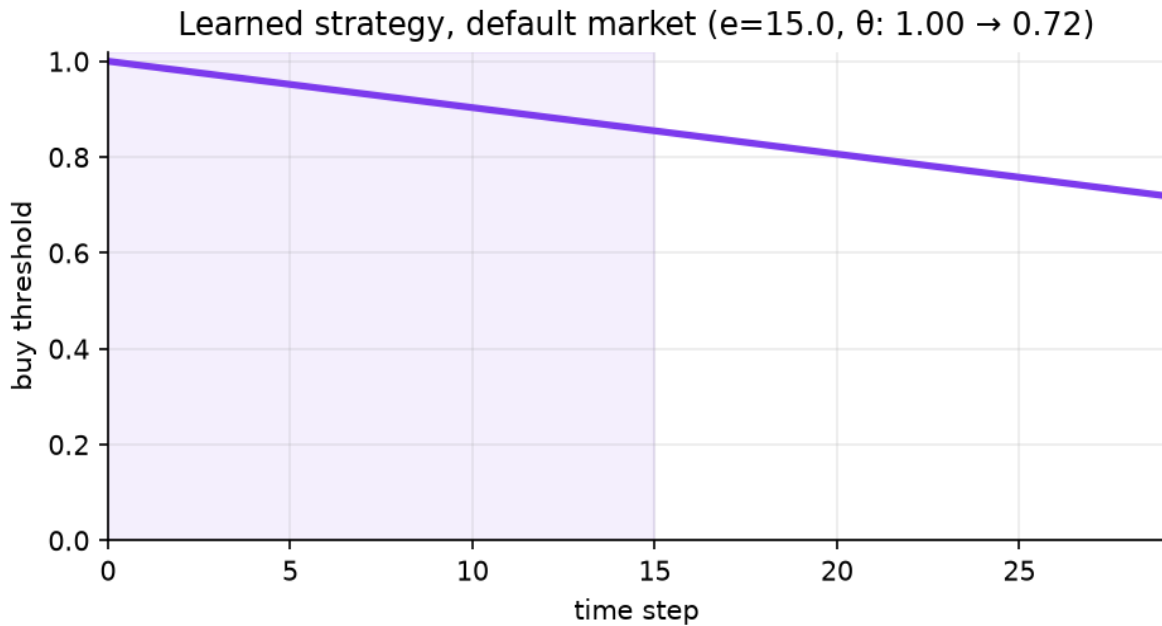


Figure 1 — A learned threshold schedule (default market: $N=8$, $M=60$, $T=30$, $stock=2$, $w=0.5$). The agent browses without buying while the bar is above what it has seen, then commits the first time a shoe clears the falling bar.

2.4 Baselines

We compare against three fixed heuristics: the **37% rule** (browse $\lceil T/e \rceil$ turns, then take the first inspected style beating the best seen while browsing — the cardinal analog of the classic rule, adapted because inspection itself consumes the budget); **fixed-0.7** (no browsing, constant bar at 0.7); and **random** (each turn, inspect then buy the best seen so far with probability 0.5).

3 Learning by self-play evolution

All N agents share one parameter vector (symmetric self-play). Fitness is mean payoff over K market episodes, evaluated with *common random numbers* — every candidate faces identical markets and acting orders within a generation, so selection sees signal, not luck. We evolve a population of 100 strategies for 10 generations (top-12 elites, Gaussian mutation), then re-evaluate the winner and all baselines on 2,000 fresh markets with paired randomness. Training seeds are recorded; re-running reproduces every number bit-for-bit. Figure 9 shows typical convergence: the population is near its final fitness by generation 7.

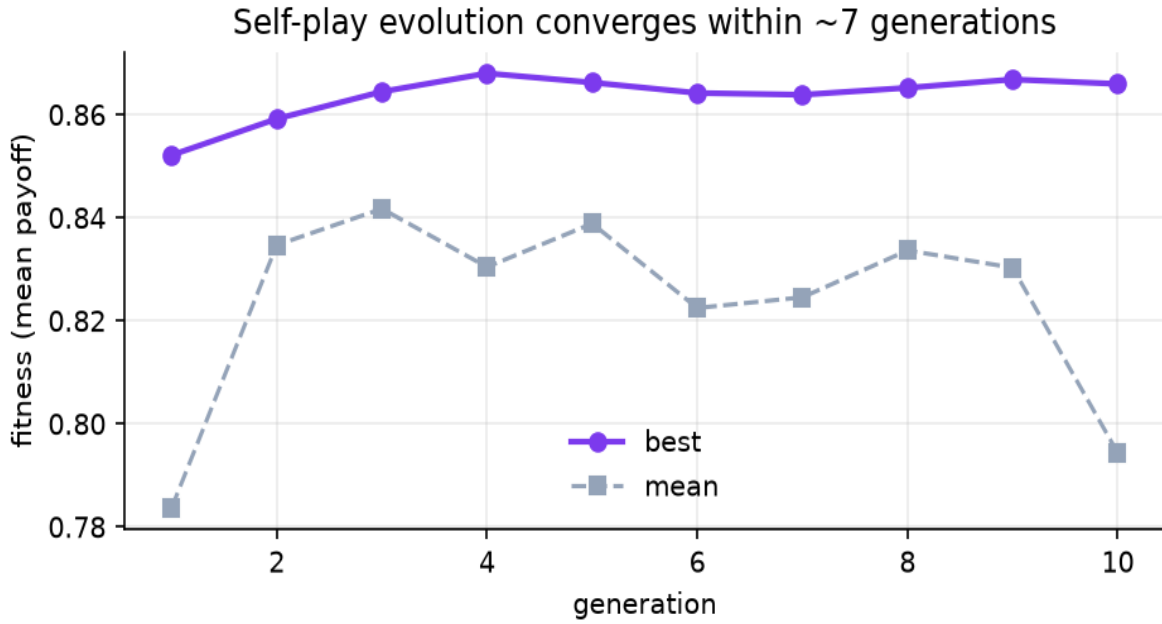


Figure 9 — Evolution of strategy fitness under self-play (default market). Selection converges within about seven generations.

Two sanity checks validate the setup. With **no competition** ($N=1$) the learner browses the maximum allowed ($e=15=T/2$, binding), runs θ : $1.00 \rightarrow 0.87$, and scores 0.955 — patient, picky, correct. Under **maximum competition** ($N=16$, stock=1, $w=1$) it learns $e=9.7$, θ : $1.00 \rightarrow 0.58$, scoring $0.855 \approx$ the symmetric optimum for 16 agents splitting the top-16 styles. The direction is right: rivalry shortens browsing and lowers the bar.

4 Experiments

Defaults held constant unless varied: $N=8$ shoppers, $M=60$ styles, $T=30$ steps, stock=2 pairs/style, $w=0.5$, uniform taste and popularity, observable stock, no safety net. Seven sweeps (27 configurations total):

Sweep	Varied	Values
Competition	N shoppers	1, 2, 4, 8, 16
Scarcity	pairs per style	1, 2, 4, 8
Satisfaction shape	taste ~ / popularity ~	uniform, normal, beta(2,5), heavy-tail
Taste agreement	w	0.0, 0.3, 0.7, 1.0
Time budget	T ($M = 2T$)	12, 30, 60
Dating market	fixed variant	$N=M=20$, $T=20$, stock=1, $w=0.9$
Safety net	fallback buy	off vs on

Evaluation SEMs are ≤ 0.0013 throughout ($K=2,000$), so any difference ≥ 0.005 quoted below is real, not noise.

5 Results

5.1 The universal: start ultra-picky, let the bar fall

Across all 27 configurations, the learned θ_0 is **1.00** — every policy starts maximally picky. The declining threshold does the exploring: with the bar near 1.0, early inspections are effectively browse-only, and the explicit browse parameter e is nearly redundant (at $N=16$, $e=0$ vs $e=15$ give identical fitness, 0.8593 vs 0.8595). The action is all in the *closing bar* θ_1 — the single adaptive dial that encodes competition, scarcity, rarity, and deadlines.

5.2 Competition: headcount matters less than you think

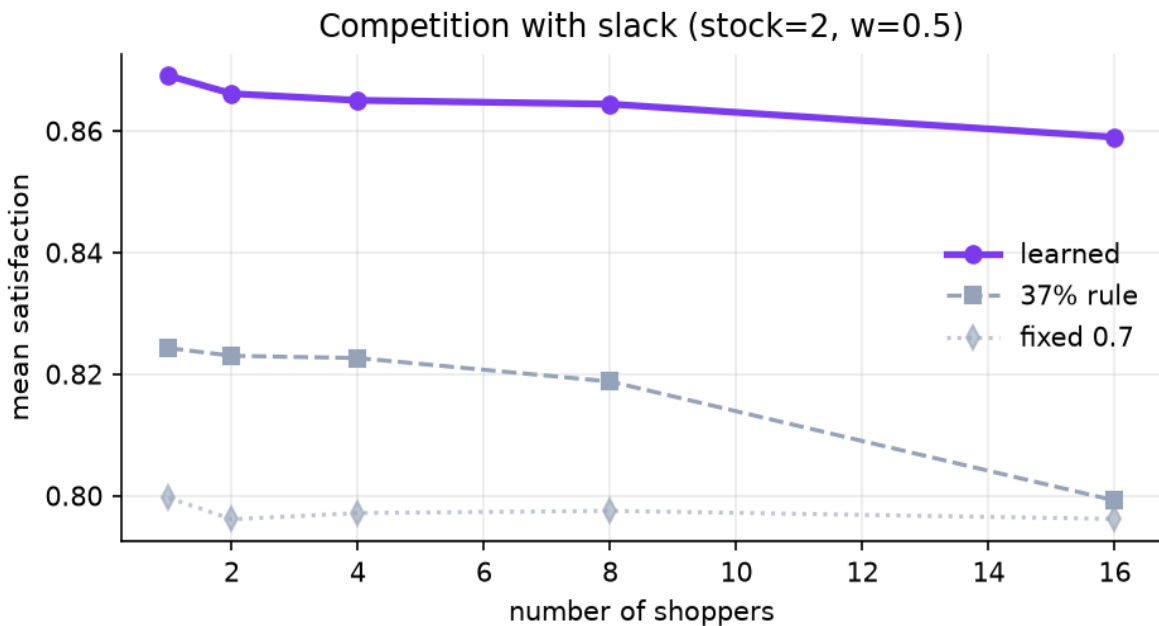


Figure 2 — Growing the crowd from 1 to 16 shoppers costs only 0.01 of welfare when stock has slack and half the value is personal taste. The learned policy beats the 37% rule by ≈ 0.05 throughout.

With stock=2 and $w=0.5$, going from 1 to 16 shoppers costs just $0.869 \rightarrow 0.859$: rivalry is diffuse because half of every shoe's value is idiosyncratic — your rivals mostly want *different* shoes than you do. The learned policy beats the 37% rule by about 0.05 at every crowd size. Headcount alone is a weak predictor of how a market feels; Sections 5.3–5.5 show what actually bites.

5.3 Scarcity: rigid rules break when the shelf empties



Figure 3 — As pairs-per-style fall from 8 to 1, the learned policy's edge over the 37% rule doubles (0.04 → 0.086).

At 1 pair per style the learned-vs-37% gap doubles to **0.086**. The mechanism is crisp: the 37% rule's browse phase records a reference shoe — and a rival buys exactly that shoe while the agent is still "exploring." After that, nothing ever clears the bar, and the agent walks out empty-handed. The learned falling bar never anchors on a vanished option; it simply keeps declining until something available clears it. Any heuristic with a fixed reference point is fragile under depletion; any heuristic with a falling bar is robust.

5.4 The shape of satisfaction sets the bar

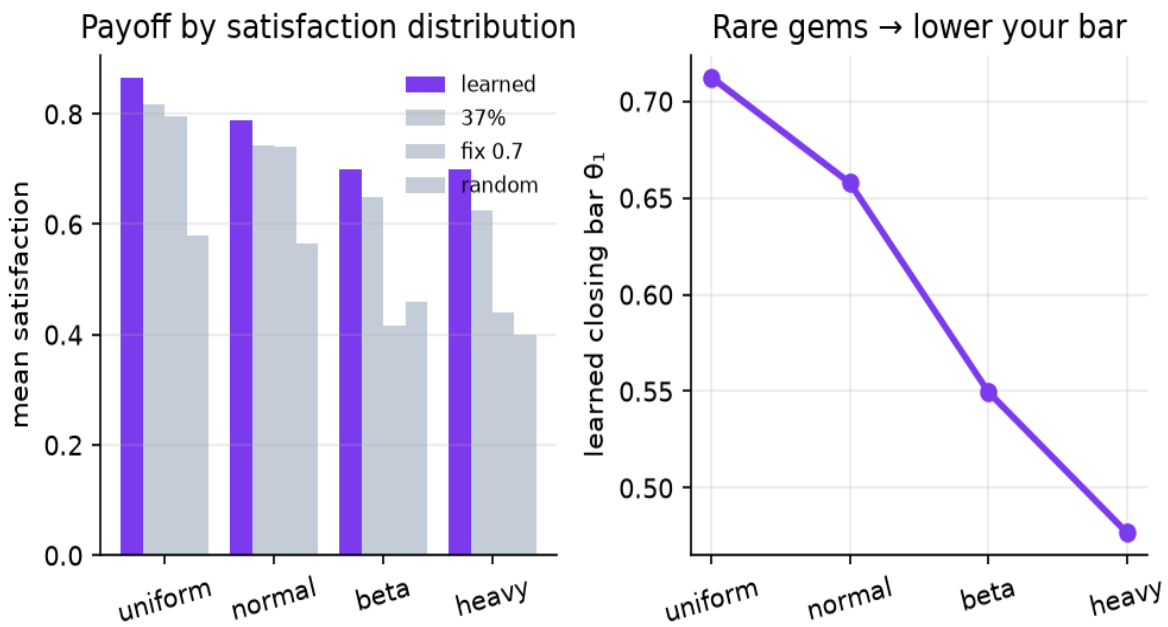


Figure 4 — Left: payoff by taste distribution. Right: the learned closing bar falls as great shoes get rarer (0.71 → 0.48). Under beta taste, fixed-0.7 (0.42) loses to random buying (0.46).

When great shoes are rare — beta(2,5) or heavy-tailed taste — the learned closing bar drops hard, to **0.55 / 0.48** from 0.71 under uniform taste. The logic: waiting for a 0.9 that never comes means going home with 0; a mediocre shoe in hand beats a fantasy. Fixed heuristics cannot adapt and are crushed: fixed-0.7 collapses to 0.42 under beta taste — *worse than buying at random* (0.46), because a 0.7 bar is almost never cleared. Under heavy tails the learned edge over the 37% rule is largest (+0.074): patience for the rare standout *plus* willingness to eventually settle is exactly what a declining bar expresses. Popularity-distribution sweeps mirror the taste sweeps almost exactly, as the $v = 0.5 \cdot p + 0.5 \cdot t$ symmetry predicts — a good internal consistency check.

5.5 Taste agreement is U-shaped — the surprise

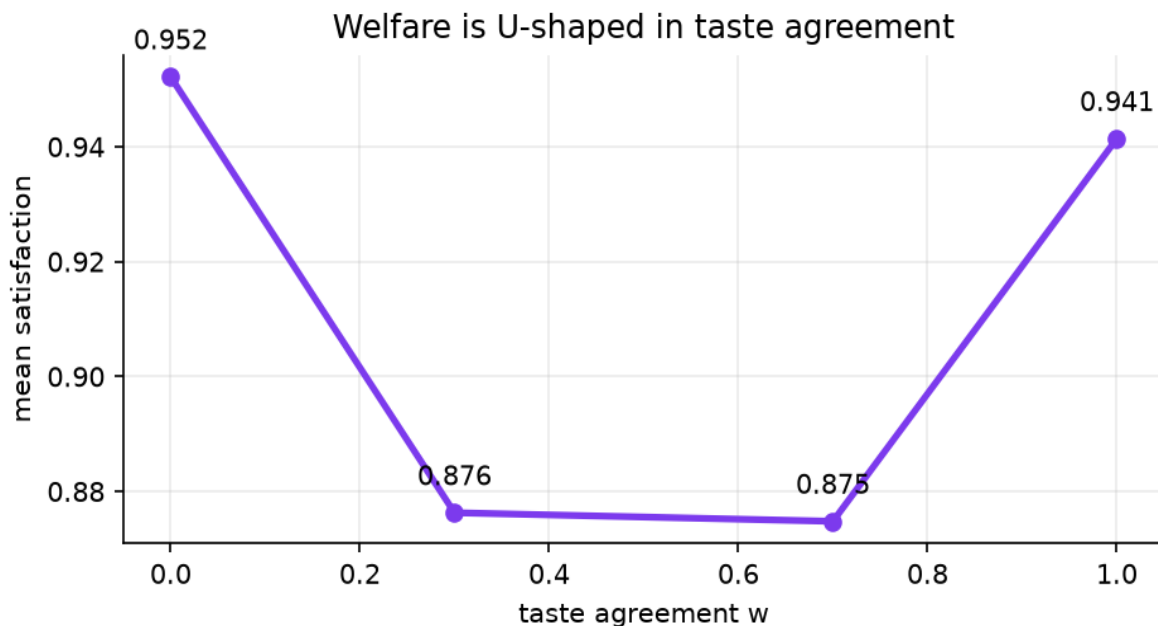


Figure 5 — Welfare vs taste agreement. No agreement (0.952) and full agreement (0.941) both beat partial agreement (≈ 0.875).

Welfare is **not** monotonic in agreement. At $w=0$ (pure personal taste) agents score 0.952 and can stay picky ($\theta_1=0.83$ — nobody competes for *your* favorites). At $w=0.3$ – 0.7 they score only ≈ 0.875 . At $w=1$ (pure popularity contest) they rebound to 0.941. Why? At $w=0.5$ you fight over shoes that are merely mediocre *for you*; the rivalry is real but the prize is diluted. At $w=1$ the fight is at least over objectively-the-best shoes, so the top styles get allocated cleanly to buyers (8 agents $\times$ the top-4 styles $\times$ 2 copies ≈ 0.96 expected). *Partial agreement is worse than full agreement or none* — the worst market is one where everyone sort of wants the same things but nobody fully agrees what they are.

5.6 Less time: commit earlier to good-enough

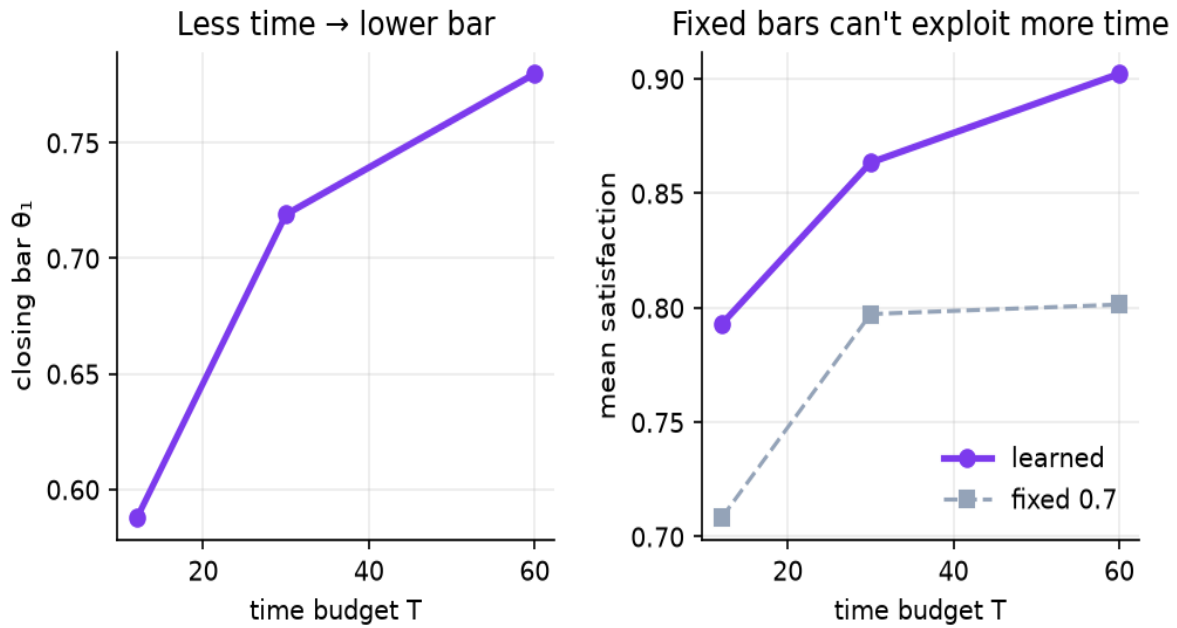


Figure 6 — Left: shorter deadlines lower the closing bar (0.59 at $T=12$ vs 0.78 at $T=60$). Right: a fixed bar cannot exploit a bigger choice set; the learned falling bar can.

Halving the budget lowers the closing bar ($\theta_1=0.59$ at $T=12$ vs 0.78 at $T=60$) and shortens browsing ($e/T \approx 0.43\text{--}0.5$ throughout — exploration scales with the budget). More time raises welfare (0.79 → 0.90: more draws, better best-seen). Notably, fixed-0.7 is flat at 0.80 across budgets: a fixed bar cannot exploit a bigger choice set, while a falling bar automatically converts extra time into extra pickiness.

5.7 The dating market: pounce, then lower your standards fast

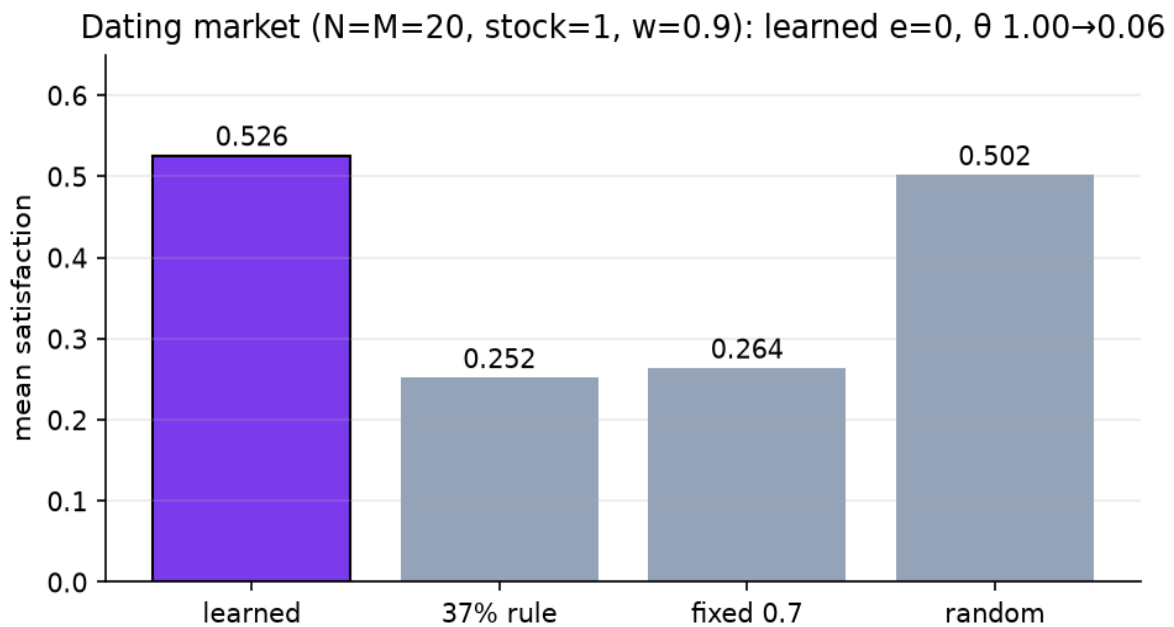


Figure 7 — Dating variant (20 suitors, 20 partners, stock=1, $w=0.9$). The learned policy (0.526) doubles the 37% rule (0.252); even random committing (0.502) beats the classic rule.

Twenty suitors, twenty partners, everyone largely agrees who's attractive ($w=0.9$), each partner pairs once — musical chairs for love. The learned strategy: $e=0$ (no browsing at all), $\theta: 1.00 \rightarrow 0.06$. In words: *pounce on love at first sight, then lower your standards fast*. Looking around while others commit is pure loss; start willing to commit only to a 10/10, but let the bar collapse toward zero as the deadline looms and the good options pair off. The fixed "play hard to get" heuristics are catastrophic (0.25–0.26): the 37% rule browses while every attractive partner pairs off, then its reference partner is gone and nothing ever clears the bar — most agents go home alone. Even *random* committing (0.50) beats the 37% rule here, because in a scramble a mediocre partner beats no partner. The learned policy beats random (+0.024, real at SEM 0.0013) through early pickiness: it still snags the occasional great match on day one.

5.8 A safety net makes you pickier

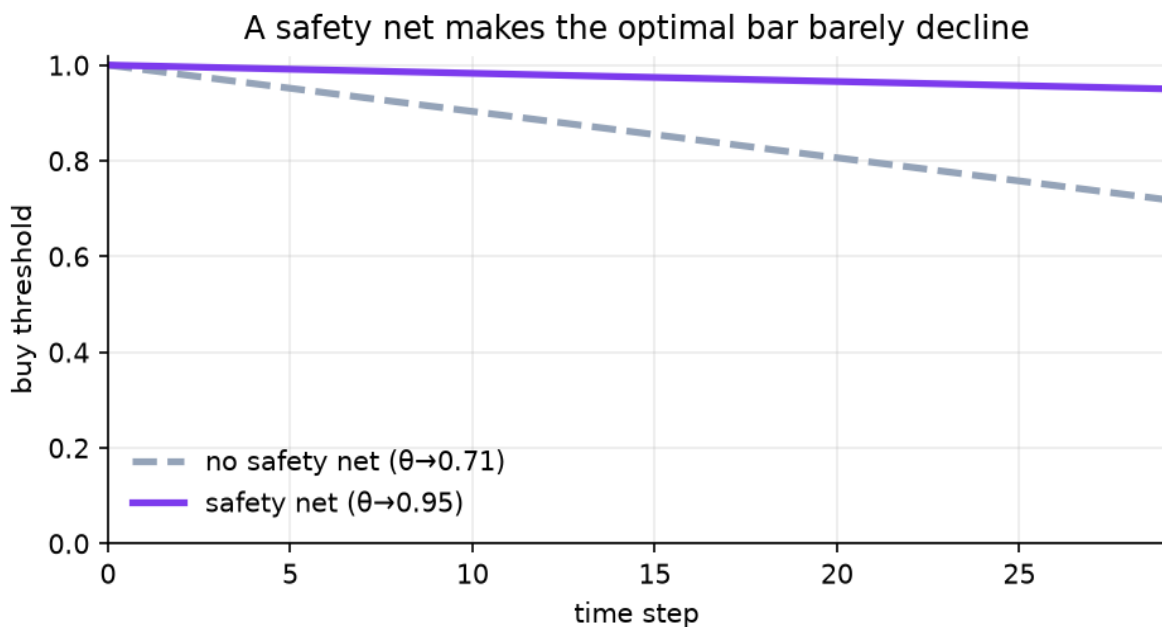


Figure 8 — With a forced best-buy at closing time, the optimal bar barely declines (1.00 $\rightarrow$ 0.95 vs 1.00 $\rightarrow$ 0.71); welfare rises 0.864 $\rightarrow$ 0.885 and nobody leaves empty-handed.

Give agents a fallback — forced to take their best inspected, available style at the deadline — and the strategy inverts: the bar barely declines (1.00 $\rightarrow$ 0.95), welfare rises to 0.885, and the empty-handed rate hits zero. With nothing to lose by waiting, hold out for excellence and let the deadline insure the downside. The real-world analog: *keep looking; if nothing turns up, order the safe pair online*. An outside option raises your reservation price — which is why people with a good fallback can afford to be picky, and people without one cannot.

6 Discussion: when to move on, when to commit

The user's question — shopping or dating, when do you move on and when do you decide "this is the one"? — gets a concrete, learned answer:

- **Commit when something clears a bar that you lower as time passes.** Move on (keep searching) while everything you've seen is below the bar. Start the bar near your maximum

realistic hope; end it near your walk-away value.

- **Set the closing bar from the market, not your dreams.** Scarce stock, short deadlines, and rare great options all push the end-bar down ($0.71 \rightarrow 0.48\text{--}0.59$); no rivalry or a safe fallback push it up ($0.83\text{--}0.95$).
- **In a scramble, never explore passively.** Browsing without the option to commit is just watching good options disappear (dating: $e=0$). Stay ready to pounce from step one, and let the *falling bar* — not a browsing phase — do your exploring.
- **Don't anchor on a specific option.** The 37% rule's failure mode is always the same: its reference shoe gets bought by someone else. Judge each option against your current bar, not against the one that got away.
- **Beware partial agreement.** Markets where everyone *sort of* wants the same things are worse than markets of pure idiosyncrasy or pure consensus. If you can, shop where tastes are personal (your edge is uncontested) or where quality is objective (the best is well-defined).

Relation to the classics: the 37% rule is the special case of our policy class with a flat-then-step bar and no depletion — optimal in its world, brittle outside it. Our agents rediscover its spirit (early pickiness) while replacing its fragile reference point with a robust falling bar. The declining reservation price also echoes optimal-search theory (e.g., McCall, 1970), here *learned* rather than derived, and extended to congested, finite-horizon choice.

7 Limitations

- **Symmetric self-play only.** All agents share one policy; we find a good symmetric profile, not a Nash equilibrium, and we do not model asymmetric exploitation (one patient agent against impatient rivals).
- The browse bound $e \leq T/2$ **binds without competition** — the unconstrained optimum wants to inspect nearly the whole budget, then take the best seen. Our policy class cannot express pure "see everything, then choose."
- **No within-trip belief updating.** $\theta(t)$ is a fixed schedule; agents don't lower the bar faster after twenty duds, as a real shopper would.
- **No prices, budgets, or search costs beyond time.** Satisfaction is the only objective; price dispersion — arguably the main driver of real shoe shopping — is absent.
- Inspection order is uniformly random (no directed search, advertising, or social influence); competition acts only through stock depletion (no bidding wars or queues). Distributions are stylized, not calibrated to real preference data.
- Exact parameter values in flat regions (especially e when $\theta_0 \approx 1$) should not be over-interpreted; the fitness landscape is flat there.

8 Conclusion

We asked when a shopper — or a dater — should stop searching and commit, and let self-play agents discover the answer under competition. They converged on a simple rule: start maximally picky, commit the first time an option clears a bar that falls as the deadline approaches, and set the

closing bar from the harshness of the market. Textbook fixed rules fail exactly where markets are hardest — scarcity, scrambles, skewed satisfaction — always by anchoring on options rivals remove. Two results deserve to travel: welfare is U-shaped in taste agreement, and a safety net is what *licenses* pickiness. The simulation engine, the interactive lab, and all 27 trained configurations are released with this paper for anyone who wants to tune the market and watch the strategies adapt.

References

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Appendix — reproducibility

Engine (engine.py), trainer (train.py), and sweep runner (run_experiments.py) are numpy-only. Training seed = 1000 + config_id; per-generation market seeds derive deterministically; final paired evaluation uses seed 2000 + config_id. Nine environment unit tests pass (test_env.py). Full per-configuration tables, learning curves, seeds, and runtimes are in FINDINGS.md and results/summary.csv.